

Abstract

We study a reflection-symmetrized Gaussian characterization arising from chi-square quadratic forms and sample variance. For a centered, variance-one random variable X with a fixed exponential moment, write M_X for its moment generating function and assume the local dominance

$$M_X(s)M_X(-s) \geq e^{s^2}.$$

We first prove an exact rigidity theorem for positive semidefinite quadratic forms of independent coordinates under the analogous coordinatewise dominance. We then establish a sharp quantitative reflection stability theorem. If

$$\Delta_\rho(X) := \sup_{|s| \leq \rho} \{ \log M_X(s) + \log M_X(-s) - s^2 \}$$

is sufficiently small, then

$$d_K(X, N(0, 1)) \leq C \sqrt{\frac{\log \log(1/\Delta_\rho(X))}{\log(1/\Delta_\rho(X))}}.$$

The proof converts the reflected MGF defect into high-order moment matching for $X - X'$, lifts this to a sub-Gaussian tail at the square-root moment scale, truncates to a bounded random variable, propagates a real-axis Gaussian product approximation into a growing complex disk, and then uses zero-freeness, a logarithmic coefficient estimate, and Esseen smoothing to recover the factor. A Laguerre-based family gives the matching lower order, showing that the double-logarithmic loss is intrinsic. We also record consequences for sample variance and explain why total-variation stability and every power-law Kolmogorov modulus fail.

Gaussian Rigidity and Sharp Reflection Stability for Chi-Square Quadratic Forms

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1 Introduction

The exact chi-square law of the sample variance is one of the classical signatures of Gaussian samples. The unrestricted converse for a fixed sample size $n \geq 3$ is, however, a long-standing characterization problem. Existing converse results impose additional structure, for example symmetry or infinite divisibility; see [4, 6, 8]. Our purpose is not to resolve that unrestricted problem. Instead, we identify a natural local symmetrized-MGF condition under which the converse becomes rigid and, more importantly, admits a sharp quantitative theory.

For a centered variance-one random variable X , let

$$M_X(s) = \mathbb{E}e^{sX}$$

whenever finite, and introduce the reflected MGF defect

$$\Delta_\rho(X) := \sup_{|s| \leq \rho} (\log M_X(s) + \log M_X(-s) - s^2). \quad (1)$$

We work under the one-sided local dominance

$$M_X(s)M_X(-s) \geq e^{s^2}, \quad |s| \leq \rho, \quad (2)$$

and a uniform exponential-integrability assumption

$$\mathbb{E}e^{2\tau|X|} \leq K. \quad (3)$$

The dominance is invisible to odd cumulants at first order: if X' is an independent copy of X , then

$$M_{X-X'}(s) = M_X(s)M_X(-s),$$

so the reflection removes the Fourier phase. The central issue is therefore a quantitative form of phase recovery compatible with probability positivity.

Our main result shows that this phase loss costs exactly a double logarithm. For sufficiently small $\Delta_\rho(X)$,

$$d_K(X, \mathcal{N}(0, 1)) \leq C_{\rho, \tau, K} \sqrt{\frac{\log \log(1/\Delta_\rho(X))}{\log(1/\Delta_\rho(X))}}. \quad (4)$$

A Laguerre family gives the reverse order, so the scale in (4) is optimal up to multiplicative constants.

1.1 Why the rate differs from classical Cramér stability

Cramér’s theorem and its stability theory concern decompositions of a nearly Gaussian convolution; see [1, 2, 9]. In the equal-factor convolution $F * F$, the Fourier transform is ϕ_F^2 , and the phase is retained. In the reflected convolution $F * \widehat{F}$, by contrast, the Fourier transform is $|\phi_F|^2$, and all phase information disappears. Our assumptions recover rigidity through a different mechanism: high-order reflected moment matching yields a square-root scale tail, truncation makes the factor entire, and a two-constants argument converts a real-axis product estimate into a growing zero-free disk for the truncated factor. The factor can then be Gaussianized through the logarithm of its characteristic function.

This mechanism is adjacent to quantitative Marcinkiewicz theorems based on zero-free regions, such as [3, 5], but the logical direction is different: a growing zero-free disk is not assumed here; it is derived from the reflection defect after truncation.

1.2 Contributions

The paper contains four main contributions.

- (i) An exact rigidity theorem for positive-semidefinite chi-square quadratic forms of independent coordinates under coordinatewise local symmetrized-MGF dominance.
- (ii) The sharp reflection-stability estimate (4) under the fixed exponential-integrability condition (3).
- (iii) A transfer from radial/sample-variance transform discrepancy to the reflected defect, giving a quantitative sample-variance corollary.
- (iv) Sharpness and obstruction results: a Laguerre family matching (4), failure of every power-law Kolmogorov modulus, and failure of total-variation stability.

1.3 Proof architecture

Set $Y = X - X'$ and $L = \log(1/\Delta_\rho(X))$. The proof chooses

$$m \asymp \frac{L}{\log L}.$$

Five lemmas drive the argument. First, a Chebyshev–Bernstein interpolation argument converts the small real-axis reflected MGF defect into $2m$ -th order moment matching for Y at accuracy $e^{-Am \log m}$. Second, the $2m$ -th moment bound for Y lifts by median symmetrization to an exponentially small tail for X at radius $N \asymp \sqrt{m}$. Third, after conditioning on $|X| \leq N$, the reflected characteristic function remains exponentially close to the Gaussian product on a real interval of radius $R \asymp \sqrt{m}$. Fourth, a two-constants theorem propagates this approximation from the real diameter to a complex disk; relative closeness to e^{-z^2} gives a zero-free disk for the truncated factor. Fifth, Carathéodory’s coefficient inequality applied to the analytic logarithm of the factor yields a quadratic expansion with remainder $O(t^3/R)$, and Esseen smoothing gives $O(1/R) = O(m^{-1/2})$.

1.4 Literature boundary

The classical fixed- n sample-variance converse without additional structural hypotheses is not claimed here. The 2017 work of Ejsmont and Lehner explicitly records that classical problem as open [4]; Ruben’s earlier result assumes symmetry [8], while Golikova

and Kruglov treat infinitely divisible variables [6]. Our theorem instead isolates local symmetrized-MGF dominance as a sufficient rigidity condition and determines the optimal stability modulus within that class.

2 Main results

Definition 2.1 (Reflected defect). *Let X be centered with $\mathbb{E}X^2 = 1$, and suppose M_X is finite on $[-\rho, \rho]$. Define*

$$\Delta_\rho(X) = \sup_{|s| \leq \rho} \{ \log M_X(s) + \log M_X(-s) - s^2 \}.$$

Whenever we invoke the stability theorem, we additionally assume the expression in braces is nonnegative on $[-\rho, \rho]$.

Theorem 2.2 (Exact PSD rigidity). *Let $X = (X_1, \dots, X_d)$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\text{Var}(X_i) = \sigma_i^2$. Let $A = A^\top \succeq 0$ have rank $r \geq 1$, and suppose*

$$X^\top A X \sim \chi_r^2.$$

For every active coordinate i with $A_{ii} > 0$, assume that $M_i(s) = \mathbb{E}e^{sX_i}$ is finite in a neighborhood of the origin and

$$M_i(s)M_i(-s) \geq e^{\sigma_i^2 s^2}$$

there. Then every active X_i is Gaussian, $X_i \sim \mathcal{N}(0, \sigma_i^2)$.

Theorem 2.3 (Sharp reflection stability). *Fix $\rho, \tau, K > 0$. There exist constants $C < \infty$ and $\Delta_0 \in (0, e^{-e})$, depending only on (ρ, τ, K) , such that the following holds. If*

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \mathbb{E}e^{2\tau|X|} \leq K,$$

and

$$0 \leq \log M_X(s) + \log M_X(-s) - s^2 \leq \Delta, \quad |s| \leq \rho,$$

for some $0 < \Delta \leq \Delta_0$, then

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\Delta)}{\log(1/\Delta)}}.$$

Corollary 2.4 (Sample-variance stability; schematic statement). *Fix $n \geq 3$ and the parameters entering the radial-transform discrepancy. Under the normalization $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, the uniform exponential-integrability assumption, and the local symmetrized-MGF dominance, if*

$$\mathfrak{d}_{n-1, \tau}(Q_n, \chi_{n-1}^2) \leq \varepsilon,$$

then, for all sufficiently small ε ,

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\varepsilon)}{\log(1/\varepsilon)}}.$$

AUDIT GATE. Insert the exact definition of $\mathfrak{d}_{n-1, \tau}$ from the project source. The handoff document states this transfer but does not reproduce the metric definition, so it must not be reconstructed from memory.

Theorem 2.5 (Sharpness). *There are constants $\rho, \tau, K, c, C > 0$ and centered variance-one random variables X_m such that the same uniform exponential-integrability and local dominance assumptions hold, $\Delta_m := \Delta_\rho(X_m) \downarrow 0$, and*

$$\log \frac{1}{\Delta_m} = m \log m + O(m), \quad d_K(X_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}.$$

Consequently

$$d_K(X_m, \mathcal{N}(0, 1)) \geq c' \sqrt{\frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)}}$$

for all sufficiently large m .

AUDIT GATE. The Imekraz–Robert–Thomann Laguerre estimate needed for the uniform bound has been independently checked. Before submission, re-audit the remaining algebra, normalization, dominance, and CDF-separation steps of the Laguerre construction from the original project records.

Proposition 2.6 (Strong-distance obstruction). *There is no uniform total-variation stability theorem under the assumptions of theorem 2.3: a standardized Poisson lattice family can satisfy a fixed exponential-integrability bound and have reflected defect tending to zero while its total-variation distance from every absolutely continuous Gaussian law equals one.*

Proposition 2.7 (No power-law Kolmogorov modulus). *For every $\alpha > 0$ there is no constant C_α for which*

$$d_K(X, \mathcal{N}(0, 1)) \leq C_\alpha \Delta_\rho(X)^\alpha$$

holds uniformly under the hypotheses of theorem 2.3.

3 Exact rigidity for PSD quadratic forms

This section proves theorem 2.2. We record the proof in a form that separates the geometric identity from the one-dimensional Cramér step.

Let $A = B^\top B$, with $B \in \mathbb{R}^{r \times d}$ of rank r , and write b_i for the i th column of B . The chi-square identity implies the exact spherical transform identity obtained by averaging the Laplace transform of BX over S^{r-1} . For a sufficiently small scalar parameter t , introduce the Gaussian-normalized log-MGF defect

$$H_t(v) := \sum_{i=1}^d \left[\log M_i(t \langle b_i, v \rangle) - \frac{\sigma_i^2 t^2 \langle b_i, v \rangle^2}{2} \right].$$

The spherical identity and Jensen yield

$$\mathbb{E}_{V \sim \text{Unif}(S^{r-1})} H_t(V) \leq 0.$$

Coordinatewise symmetrized-MGF dominance, combined with the trace/variance normalization forced by $X^\top A X \sim \chi_r^2$, yields the reverse inequality. Hence the spherical mean vanishes. Equality in the nonnegative symmetrized defect then implies, for every active i and all small s ,

$$M_i(s)M_i(-s) = e^{\sigma_i^2 s^2}.$$

Thus $X_i - X'_i$ is Gaussian. Lévy–Cramér gives X_i Gaussian.

Remark 3.1. When $r = 1$, the sphere consists only of two points. No continuous support argument is needed: varying the scalar parameter t in a neighborhood of zero still produces a full interval of arguments for each active coordinate.

AUDIT GATE. For the submission version, insert the exact spherical transform identity and the trace normalization as displayed equations. These were established in the project handoff but should be copied from the original proof record rather than reconstructed imprecisely.

4 From radial discrepancy to a pointwise reflected defect

For the sample-variance application define

$$D(s) := \frac{1}{2} \log \frac{M_X(s)M_X(-s)}{e^{s^2}} \geq 0.$$

The radial-transform comparison yields an integrated spherical bound of the form

$$n \mathbb{E} D \left(\tau \sqrt{\frac{n-1}{n}} V_1 \right) \leq \log(1 + \varepsilon), \quad (5)$$

where V_1 is the first coordinate of a uniform point on S^{n-2} . Its density is

$$f_{V_1}(v) = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma((n-2)/2)} (1 - v^2)^{(n-4)/2}, \quad -1 < v < 1.$$

For $n = 3$ this is the arcsine density.

The exponential-integrability assumption gives the uniform derivative bound

$$|D'(s)| \leq \sqrt{K} + |s|.$$

On every strict interior interval $|s| \leq \rho' < \tau \sqrt{(n-1)/n}$, the density of the spherical coordinate is bounded below. A nonnegative Lipschitz tent argument applied to (5) therefore yields

$$\sup_{|s| \leq \rho'} D(s) \leq C \sqrt{\varepsilon} \quad (6)$$

for small ε , with C depending only on the fixed parameters.

Remark 4.1. In the tent inversion one must solve in the direction $\eta \geq 2m_h I$ (equivalently $y = \eta/(2m_h)$ in the project notation); reversing this inequality gives the wrong pointwise bound.

5 Sharp reflection stability

Throughout this section let X' be an independent copy of X , set

$$Y = X - X', \quad G \sim \mathcal{N}(0, 2),$$

and define

$$F(z) := M_Y(z) - e^{z^2}, \quad L := \log(1/\Delta).$$

We prove theorem 2.3 through five lemmas.

5.1 High-order reflected moment transfer

Lemma 5.1 (Local defect to high-order moments). *Fix ρ, τ, K . For every $A > 0$ there is $c_0 = c_0(A, \rho, \tau, K) > 0$ such that, with*

$$m = \left\lfloor c_0 \frac{L}{\log L} \right\rfloor,$$

all sufficiently small Δ satisfy

$$\left| \mathbb{E}Y^k - \mathbb{E}G^k \right| \leq e^{-Am \log m}, \quad 0 \leq k \leq 2m.$$

In particular,

$$\mathbb{E}|Y|^{2m} \leq (Cm)^m.$$

Proof skeleton. Choose a fixed $0 < \rho_0 < \min\{\rho, \tau\}$ and a Bernstein ellipse $\mathcal{E}$ around $[-\rho_0, \rho_0]$ contained in the exponential-moment strip. The defect assumption gives

$$\sup_{|s| \leq \rho_0} |F(s)| \leq C\Delta,$$

while exponential integrability gives $\sup_{z \in \mathcal{E}} |F(z)| \leq C$. After scaling to $[-1, 1]$, the Chebyshev coefficients a_j therefore satisfy both

$$|a_j| \leq C\Delta, \quad |a_j| \leq C\lambda^{-j}$$

for a fixed $\lambda > 1$. Splitting the differentiated Chebyshev series at $J \asymp L$ and using

$$\|T_j^{(k)}\|_{[-1,1]} \leq j^{2k}$$

gives

$$|F^{(k)}(0)| \leq C\rho_0^{-k} \Delta L^{2k+1}, \quad k = o(L).$$

For $k \leq 2m$ and $m = c_0 L / \log L$, the logarithm of the right-hand side is

$$-L + 4c_0 L + o(L).$$

Taking c_0 sufficiently small in terms of the prescribed A yields the stated $e^{-Am \log m}$ bound. Since $F^{(k)}(0) = \mathbb{E}Y^k - \mathbb{E}G^k$, the moment conclusion follows. The $2m$ -th moment bound uses

$$\mathbb{E}G^{2m} = \frac{(2m)!}{m!} \leq (Cm)^m.$$

□

5.2 Tail lifting and bounded truncation

Lemma 5.2 (Square-root tail lifting). *There are constants $C, c > 0$ with the following property. For every sufficiently large fixed B , if $N = B\sqrt{m}$, then*

$$\mathbb{P}(|X| > N) \leq Ce^{-\beta_B m},$$

where $\beta_B \rightarrow \infty$ as $B \rightarrow \infty$. If X_N denotes the conditional law $X \mid \{|X| \leq N\}$, then

$$d_{\text{TV}}(X, X_N) \leq Ce^{-\beta_B m}.$$

Moreover

$$|\mathbb{E}X_N| + |\text{Var}(X_N) - 1| \leq C(1 + N^2)e^{-cN}.$$

Proof skeleton. Let a be a median of X . Since $\mathbb{E}X^2 = 1$, $|a| \leq \sqrt{2}$. The safe symmetrization estimate is

$$\mathbb{P}(|X - a| > x) \leq 4\mathbb{P}(|X - X'| > x).$$

Combine this with Markov's inequality and theorem 5.1. The conditional-law total-variation estimate is immediate. The mean and variance shifts are controlled separately from the original exponential moment, which gives exponentially small first and second tail moments at radius N . $\square$

5.3 A growing real Gaussian-product window

Lemma 5.3 (Real-axis product approximation). *For B fixed sufficiently large, there is $a_0 = a_0(B, \rho, \tau, K) > 0$ such that, with*

$$R = a_0\sqrt{m}$$

and f_N the characteristic function of X_N ,

$$\sup_{|t| \leq 2R} \left| f_N(t)f_N(-t) - e^{-t^2} \right| \leq e^{-\gamma_1 m}$$

for some $\gamma_1 > 0$.

Proof skeleton. Use the $2m - 1$ Taylor polynomial of $\phi_Y(t)$ and the real-variable remainder

$$\left| e^{iu} - \sum_{k=0}^{2m-1} \frac{(iu)^k}{k!} \right| \leq \frac{|u|^{2m}}{(2m)!}.$$

The high-order moment mismatch from theorem 5.1 is negligible on $|t| \leq 2R$, while

$$\frac{|t|^{2m} \mathbb{E}|Y|^{2m}}{(2m)!} \leq (Ca_0^2)^m.$$

Choose a_0 small. Finally transfer from X to X_N using theorem 5.2. $\square$

5.4 Complex propagation and a zero-free factor disk

Lemma 5.4 (Growing zero-free disk). *The constants B and a_0 can be chosen so that, for $R = a_0\sqrt{m}$,*

$$f_N(z) \neq 0, \quad |z| \leq R.$$

More precisely, for some $\gamma_2 > a_0^2$,

$$\sup_{|z| \leq R} \left| f_N(z)f_N(-z) - e^{-z^2} \right| \leq e^{-\gamma_2 m}.$$

Proof skeleton. Since $|X_N| \leq N$, f_N is entire and

$$|f_N(z)| \leq e^{N|\Im z|}.$$

Hence, for

$$\mathcal{F}_N(z) := f_N(z)f_N(-z) - e^{-z^2},$$

one has

$$\log |\mathcal{F}_N(z)| \leq C(NR + R^2) \leq C(Ba_0 + a_0^2)m, \quad |z| \leq 2R.$$

On the real diameter, theorem 5.3 gives $\log |\mathcal{F}_N| \leq -\gamma_1 m$. Apply the two-constants theorem separately in the upper and lower half-disks of radius $2R$. The harmonic measure of the diameter is uniformly bounded below on the concentric half-disk of radius R . Choose first B large and then a_0 sufficiently small so that the propagated exponent γ_2 is positive and satisfies $\gamma_2 > a_0^2$. Since

$$\inf_{|z| \leq R} |e^{-z^2}| \geq e^{-R^2} = e^{-a_0^2 m},$$

relative closeness implies $f_N(z)f_N(-z) \neq 0$, hence $f_N(z) \neq 0$ on the disk. $\square$

5.5 Logarithmic Gaussianization and Esseen smoothing

Lemma 5.5 (Zero-free factor to Kolmogorov distance). *Under the conclusions of theorem 5.4,*

$$d_K(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \leq \frac{C}{R},$$

where $\mu_N = \mathbb{E}X_N$ and $\sigma_N^2 = \text{Var}(X_N)$. Consequently

$$d_K(X, \mathcal{N}(0, 1)) \leq \frac{C}{\sqrt{m}}.$$

Proof skeleton. On $|z| < R$ choose the analytic logarithm

$$\psi_N(z) = \log f_N(z), \quad \psi_N(0) = 0.$$

The bounded support gives

$$\Re \psi_N(z) = \log |f_N(z)| \leq NR.$$

Carathéodory's coefficient inequality therefore yields, writing $\psi_N(z) = \sum_{k \geq 1} a_k z^k$,

$$|a_k| \leq 2NR^{1-k}.$$

Since $a_1 = i\mu_N$ and $a_2 = -\sigma_N^2/2$, for every fixed $0 < \theta < 1$,

$$\psi_N(t) = i\mu_N t - \frac{\sigma_N^2 t^2}{2} + O\left(\frac{|t|^3}{R}\right), \quad |t| \leq \theta R,$$

because $N/R = B/a_0$ is fixed. By theorem 5.2, $\mu_N = o(R^{-1})$ and $\sigma_N^2 = 1 + o(R^{-1})$. Taking θ sufficiently small gives Gaussian damping, and hence

$$\left| f_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2} \right| \leq C \frac{|t|^3}{R} e^{-ct^2}.$$

Esseen smoothing with cutoff $T = \theta R$ gives C/R . Add the truncation and parameter-shift errors. $\square$

Proof of theorem 2.3. Apply theorems 5.1 to 5.5. Since

$$m \asymp \frac{L}{\log L}, \quad L = \log(1/\Delta),$$

we have

$$\frac{1}{\sqrt{m}} \asymp \sqrt{\frac{\log L}{L}},$$

which is the claimed rate. $\square$

6 Sample-variance stability

Let $X_1, \dots, X_n$ be i.i.d. copies of X and

$$Q_n = \sum_{i=1}^n (X_i - \bar{X})^2.$$

Under the radial-transform discrepancy used in the project, eq. (6) gives

$$\Delta_{\rho'}(X) \leq C\sqrt{\varepsilon}.$$

Applying theorem 2.3 therefore gives

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\varepsilon)}{\log(1/\varepsilon)}}$$

for sufficiently small ε ; the factors $1/2$ and fixed multiplicative constants inside the logarithms only change the implicit constant.

AUDIT GATE. Replace the schematic radial metric with the exact definition and hypotheses from the source theorem before circulating the manuscript.

7 Sharpness and obstruction results

7.1 Laguerre sharpness family

The construction uses generalized Laguerre polynomials $L_m^{(1/2)}$. The external uniform estimate required by the construction follows from Proposition 3.2 of Imekraz, Robert and Thomann [7]. With

$$\nu = 4m + 3, \quad Q_m(r) = \sqrt{r} e^{-r/2} |L_m^{(1/2)}(r)|,$$

the four-region estimate implies

$$\sup_{m \geq 0, r \geq 0} \sqrt{r} e^{-2r/3} |L_m^{(1/2)}(r)| < \infty.$$

The construction then yields a uniformly exponentially integrable family satisfying local dominance and

$$\log \frac{1}{\Delta_m} = m \log m + O(m), \quad d_K(X_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}.$$

Stirling inversion gives

$$m^{-1/2} \asymp \sqrt{\frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)}}.$$

AUDIT GATE. This subsection deliberately states only the already-audited external Laguerre bound and the project-recorded asymptotic package. Before submission, reconstruct every definition of X_m and verify normalization, positivity, local dominance, exponential integrability, defect asymptotics, and CDF separation from first principles.

7.2 Failure of power rates

A high-frequency odd Gaussian perturbation, accompanied by a small symmetric lattice compensation and variance renormalization, gives a family with fixed exponential integrability and local dominance such that

$$\Delta_n \asymp n^6 e^{-n^2}, \quad d_K(X_n, \mathcal{N}(0, 1)) \geq \frac{c}{n}.$$

Thus no bound of the form $C\Delta^\alpha$ can hold uniformly for any $\alpha > 0$.

AUDIT GATE. Insert the full construction and proof from the project record if this proposition is retained in the main paper. It may be shortened or moved to supplementary material once the Laguerre sharpness theorem is complete.

7.3 Failure of total-variation stability

For $N_\lambda \sim \text{Poisson}(\lambda)$, set

$$X_\lambda = \frac{N_\lambda - \lambda}{\sqrt{\lambda}}.$$

Then the reflected defect tends to zero at fixed local scale, while X_λ is lattice-valued and the Gaussian law is absolutely continuous, so

$$d_{\text{TV}}(X_\lambda, \mathcal{N}(0, 1)) = 1.$$

8 Discussion and open directions

The sharp reflection rate is slower than what is available in equal-factor Cramér stability because reflection erases the characteristic-function phase. The present proof recovers the phase indirectly by manufacturing a growing zero-free disk after a square-root-scale truncation. This mechanism may be useful in other factor-recovery problems in which only a modulus or autocorrelation is initially observed.

The unrestricted fixed- n sample-variance characterization remains outside the scope of the paper: our exact and quantitative results use local symmetrized-MGF dominance. Removing that assumption would require a new mechanism for controlling the odd/phase degrees of freedom.

Several exploratory descriptions—Jacobi square budgets, Toeplitz–Schur chains, and Rogers–Szegő certificates—help explain the factorial $m \log m$ scale but are not needed in the streamlined proof. They may be developed separately if they lead to an independent structural theorem.

A Parameter selection for the sharp upper bound

This appendix records a non-circular order of choices.

1. Fix the model parameters (ρ, τ, K) and an interior analytic radius ρ_0 .
2. Choose a target moment-accuracy exponent A large enough to dominate all later finite-order losses.
3. Choose $c_0 > 0$ sufficiently small so that theorem 5.1 yields $e^{-Am \log m}$ accuracy for $k \leq 2m$.

4. Choose the truncation constant B large, producing a tail exponent β_B as large as required.
5. Choose $a_0 > 0$ small enough that: (i) the characteristic-function Taylor remainder on $|t| \leq 2a_0\sqrt{m}$ is $e^{-\gamma m}$ with γ larger than the desired threshold; (ii) the complex-growth constant $C(Ba_0 + a_0^2)$ in the half-disk is sufficiently small relative to the real-axis exponent after multiplication by the harmonic-measure constant; and (iii) the propagated exponent γ_2 exceeds a_0^2 .
6. Finally choose the fixed Esseen fraction $\theta > 0$ small enough that the cubic logarithmic remainder is dominated by a fixed fraction of the quadratic Gaussian damping.

All choices depend only on the fixed model parameters, not on Δ or m .

AUDIT GATE. In the submission version, give one explicit system of inequalities among $(A, c_0, B, a_0, \gamma, \gamma_1, \gamma_2, \theta)$ and verify them in this order.

B Laguerre input and final sharpness audit

For reference, Imekraz–Robert–Thomann normalize the Laguerre function by

$$\mathcal{L}_m^\alpha(r) = a_{m,\alpha} L_m^{(\alpha)}(r) e^{-r/2} r^{\alpha/2}, \quad a_{m,\alpha} = \sqrt{\frac{m!}{\Gamma(m + \alpha + 1)}},$$

with

$$\nu = 4m + 2\alpha + 2.$$

For $\alpha = 1/2$, $\nu = 4m + 3$ and

$$Q_m(r) = \sqrt{r} e^{-r/2} |L_m^{(1/2)}(r)| = a_m^{-1} r^{1/4} |\mathcal{L}_m^{1/2}(r)|.$$

Their four-region Proposition 3.2 yields: a uniform bound in the first two regions, $O(\nu^{1/6})$ in the turning region, and exponential decay outside. Multiplication by $e^{-r/6}$ therefore gives

$$\sup_{m,r} \sqrt{r} e^{-2r/3} |L_m^{(1/2)}(r)| < \infty.$$

The final pre-submission audit must then verify the remaining internal construction chain:

1. definition and nonnegativity of the perturbed density/law;
2. exact centering and variance normalization;
3. a uniform exponential-integrability constant;
4. local symmetrized-MGF dominance;
5. defect asymptotics $\log(1/\Delta_m) = m \log m + O(m)$;
6. Kolmogorov separation $d_K(X_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}$.

No novelty or sharpness claim should be submitted before these six items have been independently rechecked.

References

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